

Rodrigo F. Grande Varg

Quantitative Economist — Data Science, Time Series & Forecasting

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Summary

Quantitative economist with 14+ years at the Central Reserve Bank of Peru (BCRP), building forecasting, nowcasting and machine learning models that feed monetary policy decisions every week. My work sits where statistical rigor meets production data: high-frequency microdata panels, automated collection pipelines, mixed-frequency models and formal out-of-sample evaluation. Delivered a 15% RMSE reduction in the inflation nowcast and a daily credit indicator published in a peer-reviewed journal. MSc in Economics and Finance (Barcelona School of Economics) and MSc in Applied Mathematics (PUCP, ranked first). I teach Machine Learning in the MSc in Artificial Intelligence at UPC.

Professional Experience

Central Reserve Bank of Peru (BCRP)

Sep 2019 – Present

Senior Price Specialist

- Built mixed-frequency and machine learning nowcasting models for inflation (XGBoost, forecast combination), achieving a 15% RMSE reduction over the previous specification.
- Own the short-term inflation forecast: two weekly forecast reports (headline and core inflation) used as direct input for monetary policy decisions.
- Developed a daily high-frequency credit indicator (Random Forest and dynamic network connectedness) to anticipate turning points in the credit cycle; published in the *Latin American Journal of Central Banking*.
- Build and curate high-frequency price microdata panels: integration of heterogeneous sources, missing-data treatment and consistency checks on daily and weekly series.
- Automated data collection with Python web scraping, using reproducible extraction, cleaning and validation pipelines.
- Built economic indicators from unstructured data (NLP) and interactive dashboards (Plotly Dash, Power BI) to communicate results to decision-makers.

Central Reserve Bank of Peru (BCRP)

Jul 2013 – Aug 2018

Economic Outlook Specialist

- Implemented leading indicators of economic activity (BCRP Working Paper 2017-01).
- Forecast macroeconomic variables and contributed to the *Inflation Report*.

Central Reserve Bank of Peru (BCRP)

May 2012 – Jun 2013

Balance of Payments Specialist

- Compiled and analyzed the financial account of the balance of payments.

Teaching

Universidad Peruana de Ciencias Aplicadas (UPC)

Mar 2026 – Present

Lecturer, Machine Learning — MSc in Artificial Intelligence: statistical learning, linear and regularized models, SVM, probabilistic models, clustering and advanced model evaluation; Python labs (scikit-learn) on reproducible modeling pipelines, cross-validation and preventing data leakage.

Universidad San Martín de Porres

Mar 2020 – Jul 2024

Lecturer, Applied Time Series and Economic Growth (ARIMA, forecasting and model evaluation in Python).

Universidad Nacional de Ingeniería

Mar 2018 – Jul 2018

Lecturer, International Economics.

Education

Barcelona School of Economics (Universitat Pompeu Fabra)

2018 – 2019

MSc in Economics and Finance, Finance Program — GPA 8.01/10

Relevant courses: *Econometric Methods, Financial Econometrics, Machine Learning for Finance, Risk Management in Banks.*

Selected Projects (github.com/Rodgrandez)

- **peru-inflation-nowcasting:** nowcasting Lima CPI inflation with U-MIDAS, Almon-MIDAS, Ridge/LASSO, XGBoost and forecast combination on daily BCRP data; expanding-window evaluation 2015–2026 and Diebold-Mariano tests.
- **lima-food-prices-pipeline:** automated daily pipeline for Lima wholesale food prices (MIDAGRI-SISAP, 2010–2026) with published data-quality checks, price-pressure indicators and a live dashboard.
- **credit-risk-pd-validation:** PD scorecard (WoE logistic), XGBoost challenger with SHAP, LGD and expected loss, plus a second-line validation report (discrimination, calibration, PSI, backtesting).
- **bank-account-fraud-detection:** fraud detection on 1M bank account applications (Feedzai BAF) with strictly time-based validation, cost- and budget-based alert decisions, calibration, a fairness audit by age, drift monitoring and a model card.

Research and Publications

- Grandez, R., & Pardo-Figueroa, R. (2026). *A high frequency indicator of credit in Peru: A Random Forests and dynamic network connectedness approach.* Latin American Journal of Central Banking. **(Peer-reviewed)**
- Grandez, R., & Pardo, R. (2025). *Nowcasting and Backcasting Credit with Mixed-Frequency Data and Machine Learning: Evidence from Peru.* SSRN Working Paper No. 5408526.
- Flores, J., & Grandez, R. (2024). Machine learning methods to forecast inflation in Peru. *Revista Moneda*, No. 200, BCRP (in Spanish).
- Pérez Forero, F., Ghurra Aguilar, O., & Grandez Vargas, R. (2017). *A Leading Indicator for the Peruvian Economic Real Activity.* BCRP Working Paper No. 2017-01.
- MSc theses: *SABR: A stochastic volatility model in practice* (BSE, awarded) and *Option pricing with Lévy processes* (PUCP).

Technical Skills

Machine learning: Supervised and unsupervised learning, XGBoost, Random Forest, SVM, regularization, clustering, applied NLP; scikit-learn, PyTorch.

Statistics and model evaluation: Out-of-sample evaluation, cross-validation, data-leakage prevention, forecast combination, error metrics (RMSE, MAE), Diebold-Mariano tests.

Time series and econometrics: ARIMA/SARIMA, VAR/VECM, cointegration (Engle-Granger, Johansen), unit-root tests (ADF, PP, KPSS), mixed-frequency models (MIDAS/U-MIDAS), HP/BK filters.

Data engineering: Python (NumPy, Pandas), SQL, MongoDB, Git; microdata panel construction, web scraping, data quality control and reproducible pipelines.

Other tools: R, MATLAB, EViews; Plotly Dash, Power BI, Tableau.

Languages: Spanish (native), English (advanced), Portuguese (basic).